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(54) **SURGICAL STAPLER WITH UNIVERSAL TIP RELOAD**

(71) Applicant: **Covidien LP**, Mansfield, MA (US)

(72) Inventors: **Pawel Abramek**, Berlin, CT (US); **Kenneth M. Cappola**, Monroe, CT (US); **Matthew J. Chowaniec**, Madison, CT (US); **Zalika Walcott**, Hamden, CT (US)

(73) Assignee: **Covidien LP**, Mansfield, MA (US)

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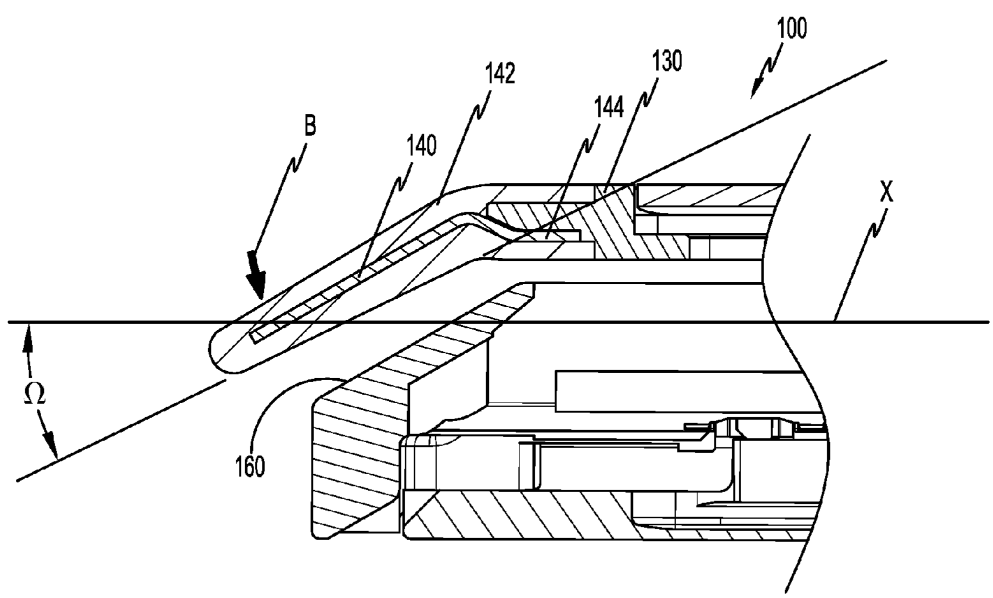
(57) **ABSTRACT**

A surgical stapler includes a tool assembly that has a cartridge assembly, an anvil assembly, and a dissector tip. The dissector tip is moveable between a first configuration aligned with a longitudinal axis of the tool assembly and a second configuration defining an acute angle with the longitudinal axis of the tool assembly.

6 Claims, 8 Drawing Sheets



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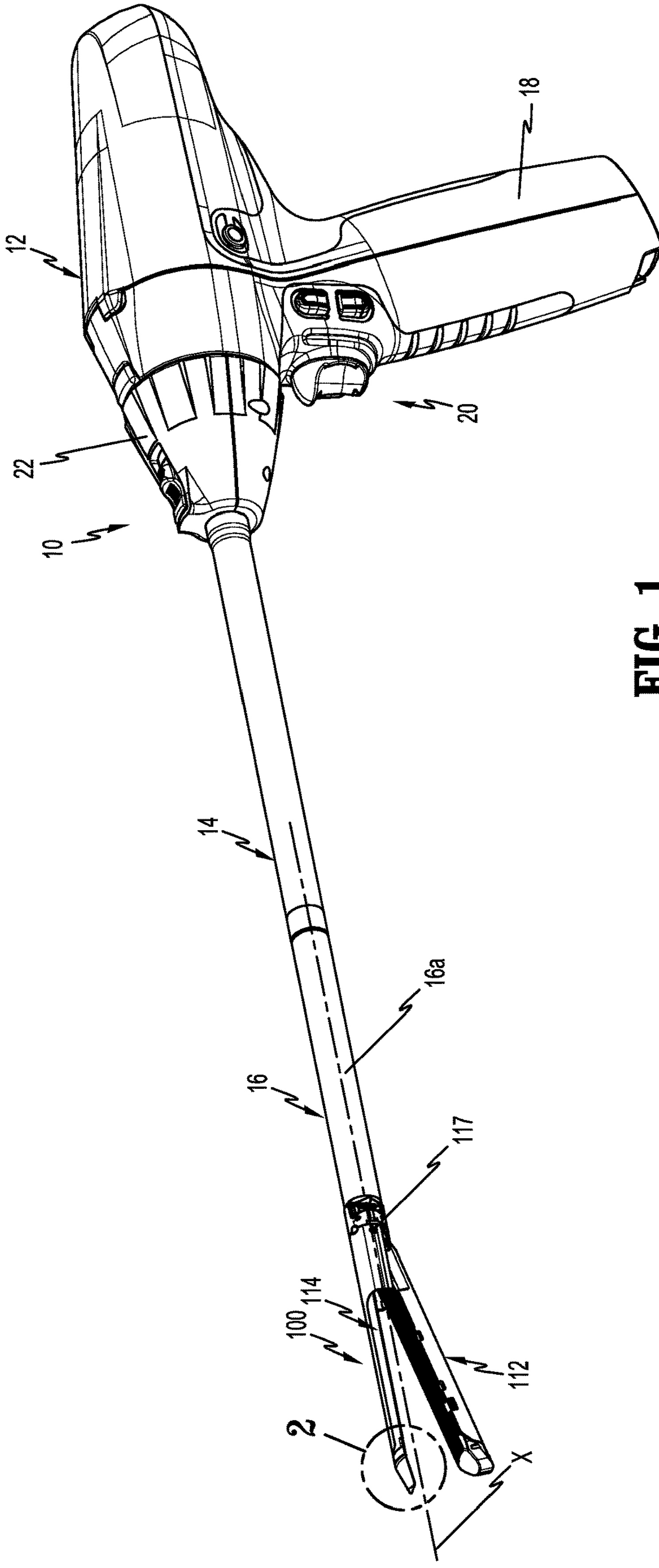
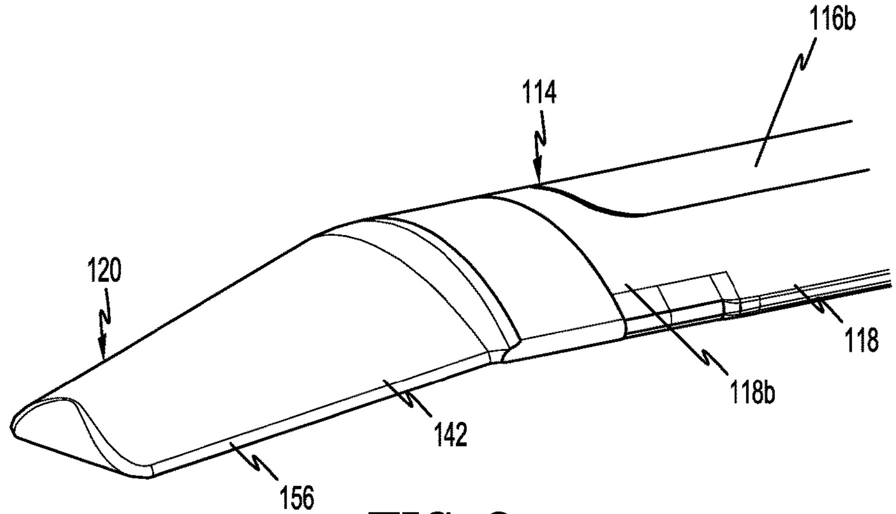


FIG. 1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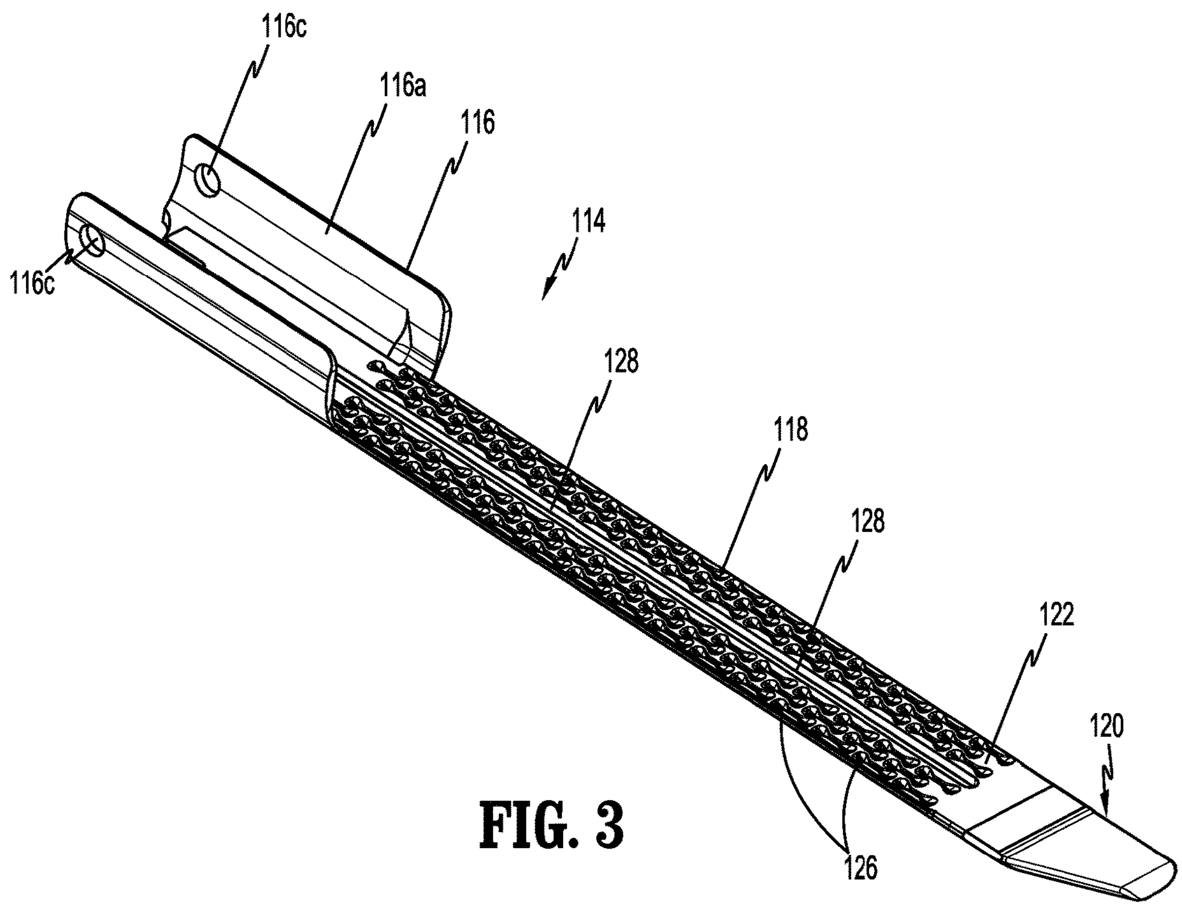
114
120
118
118b
142
156

FIG. 2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116a
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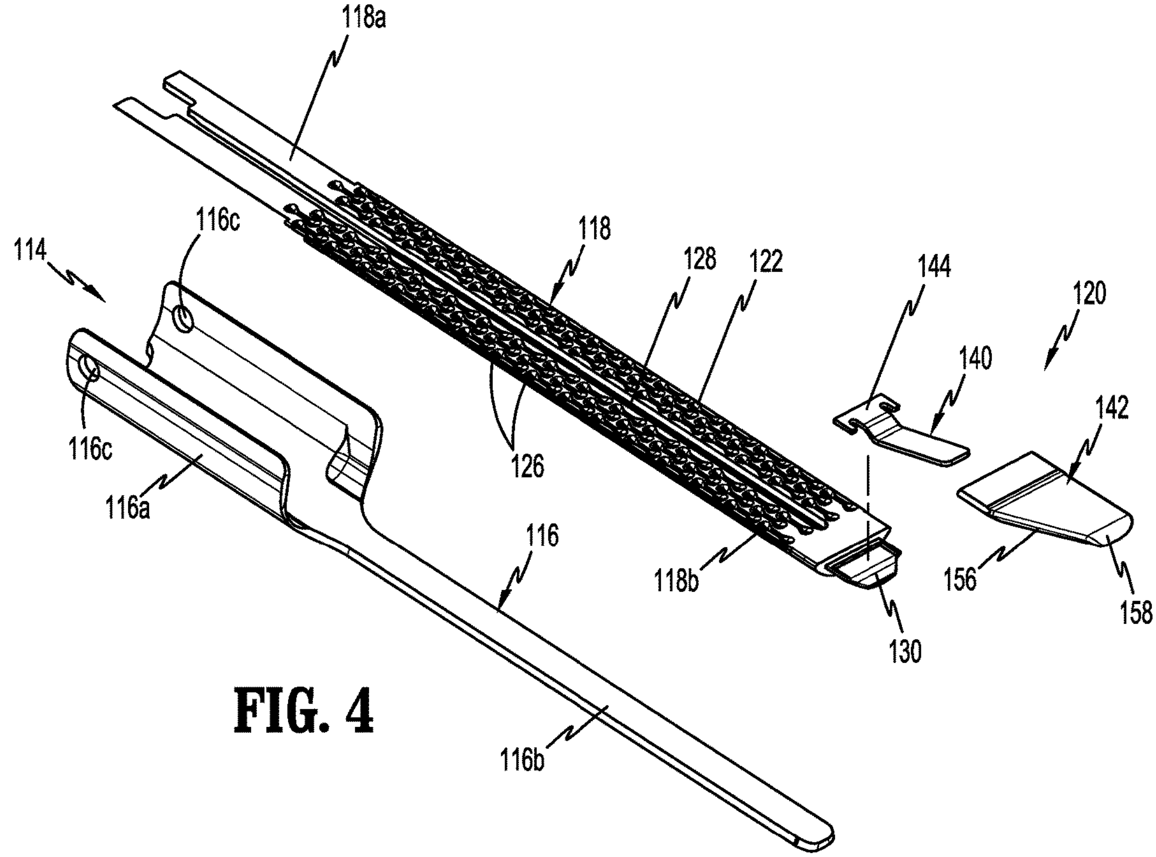
126

FIG. 3


118a
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116c
116c
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116
118b
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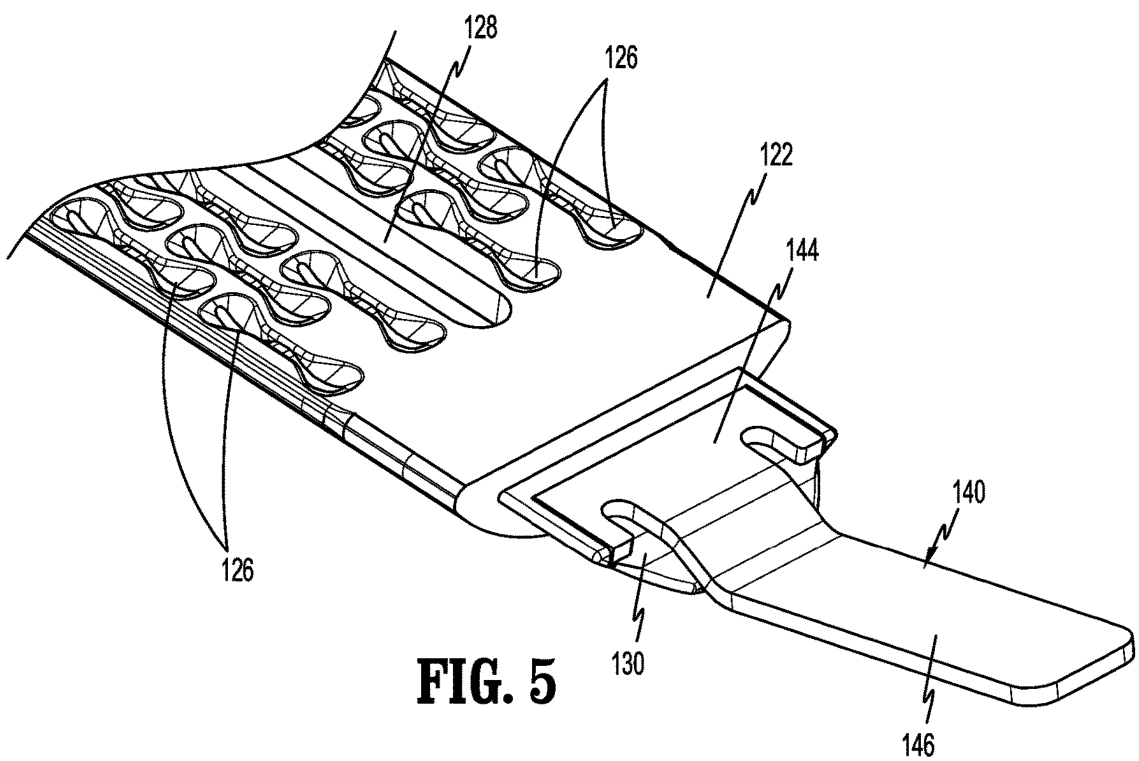
130
116b

FIG. 4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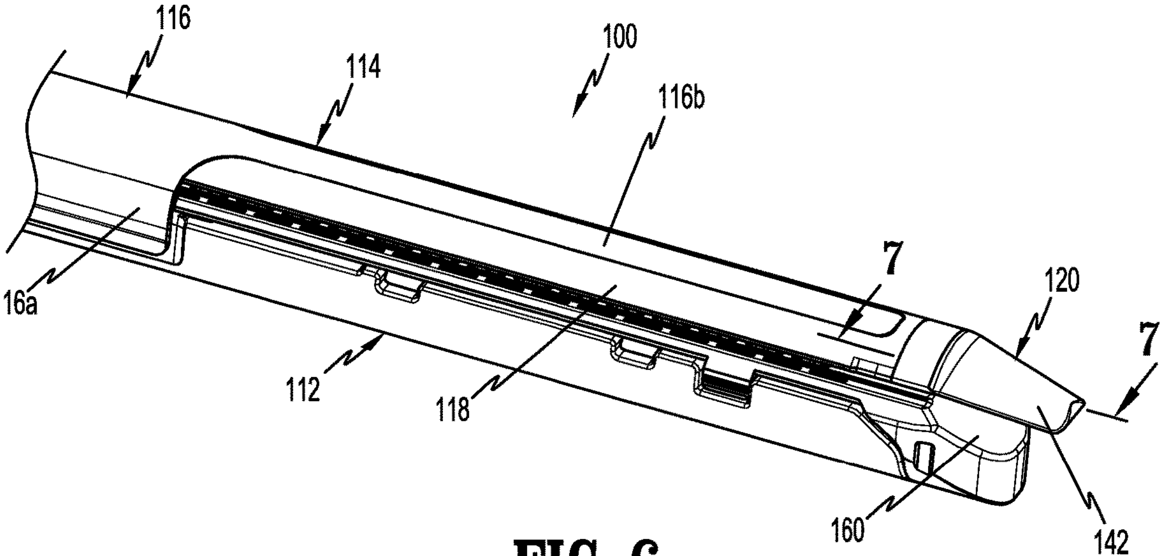
146

FIG. 5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116a
112
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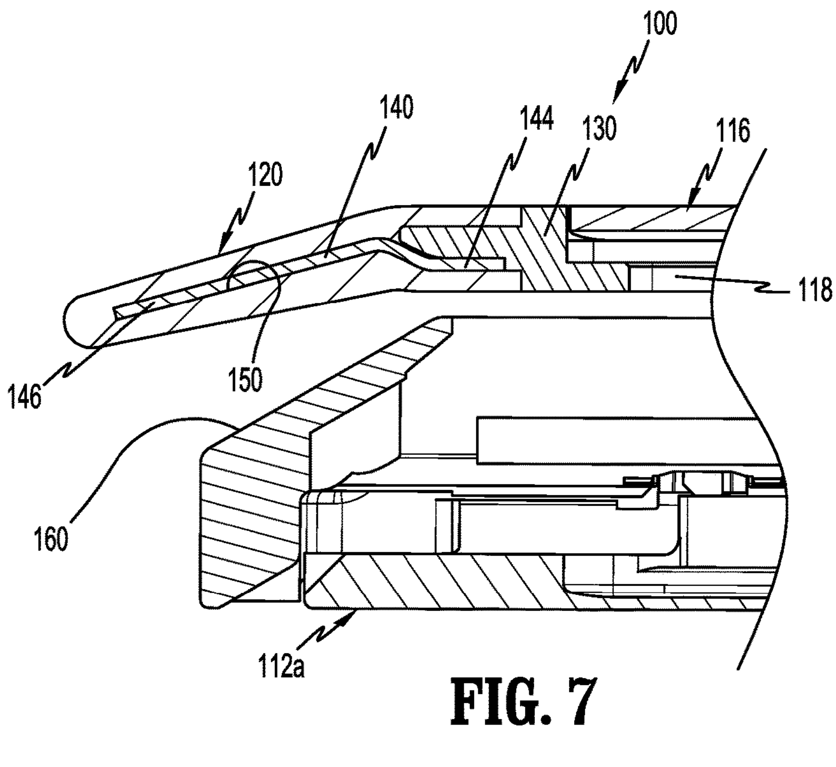
160
142

FIG. 6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116
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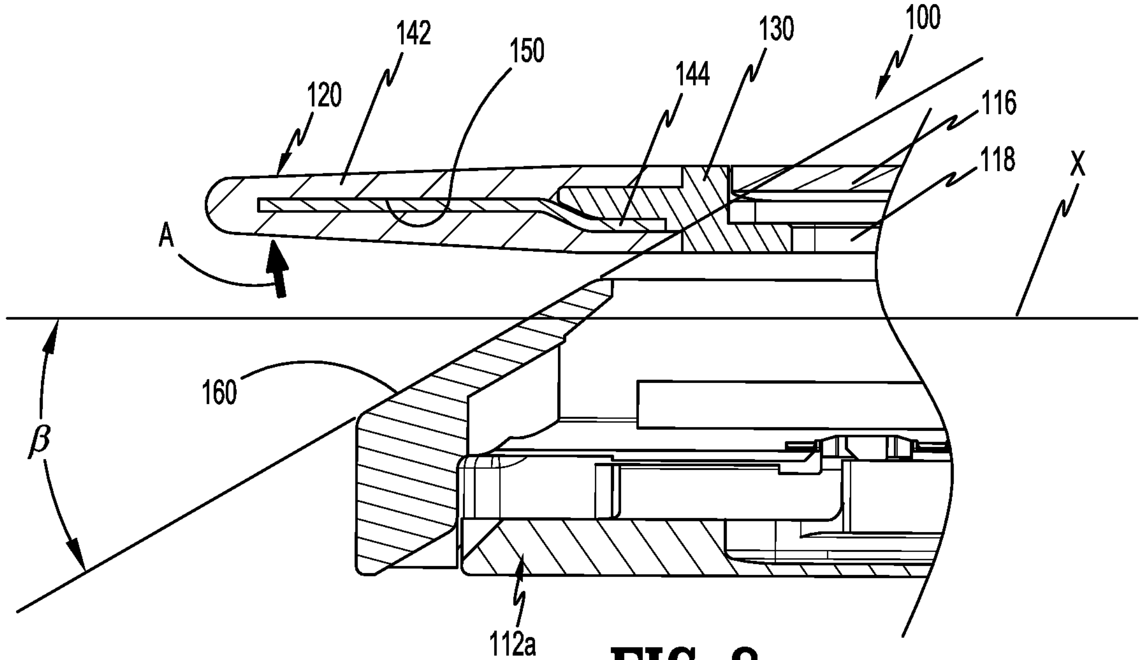
146
150
160
112a

FIG. 7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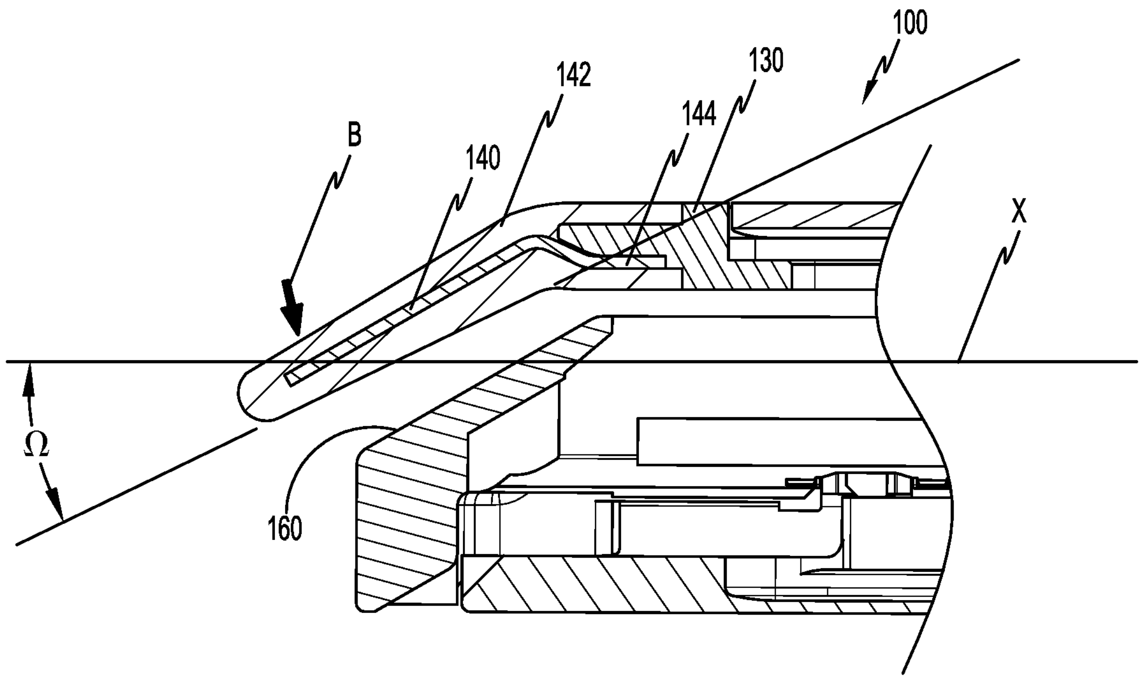
130
100
116
118
X
A
160
β
112a

FIG. 8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142
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B
144
140
X
Ω
160

FIG. 9

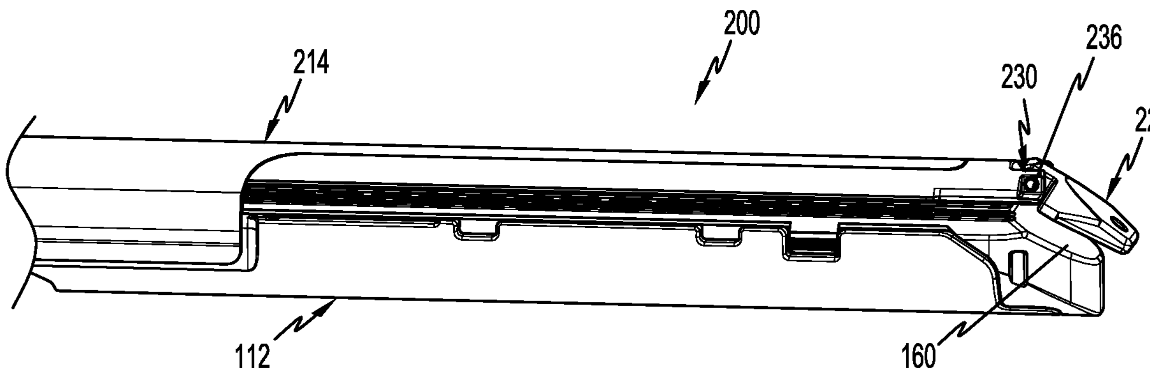
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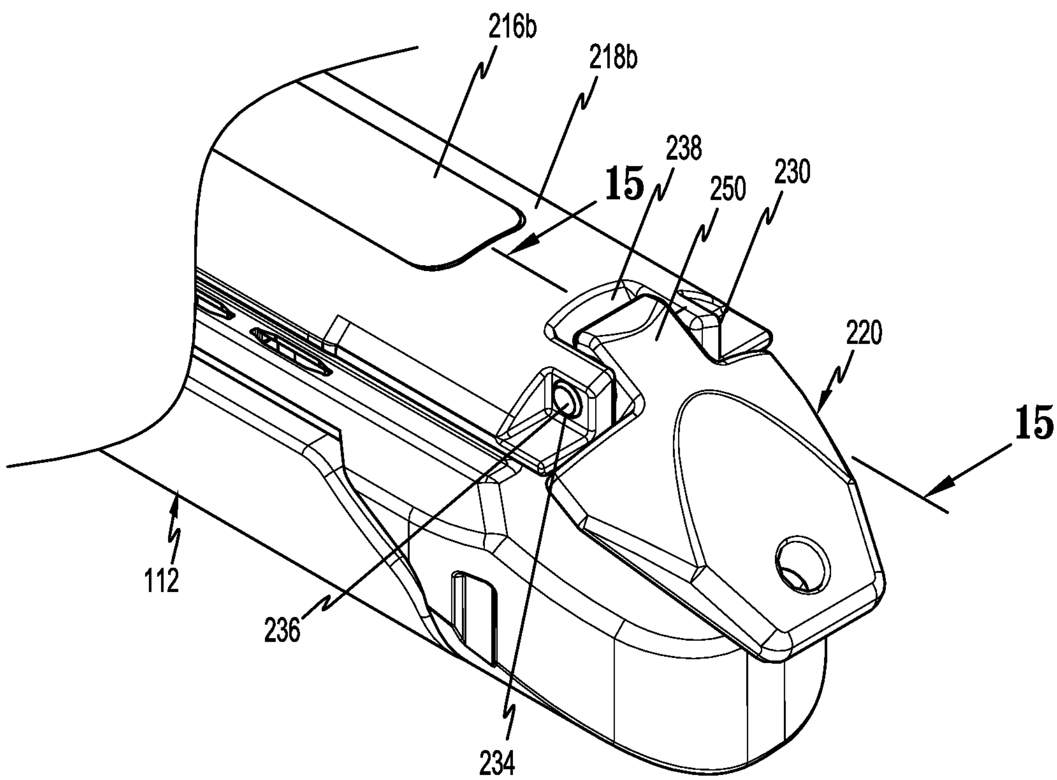
FIG. 10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236
234

FIG. 11

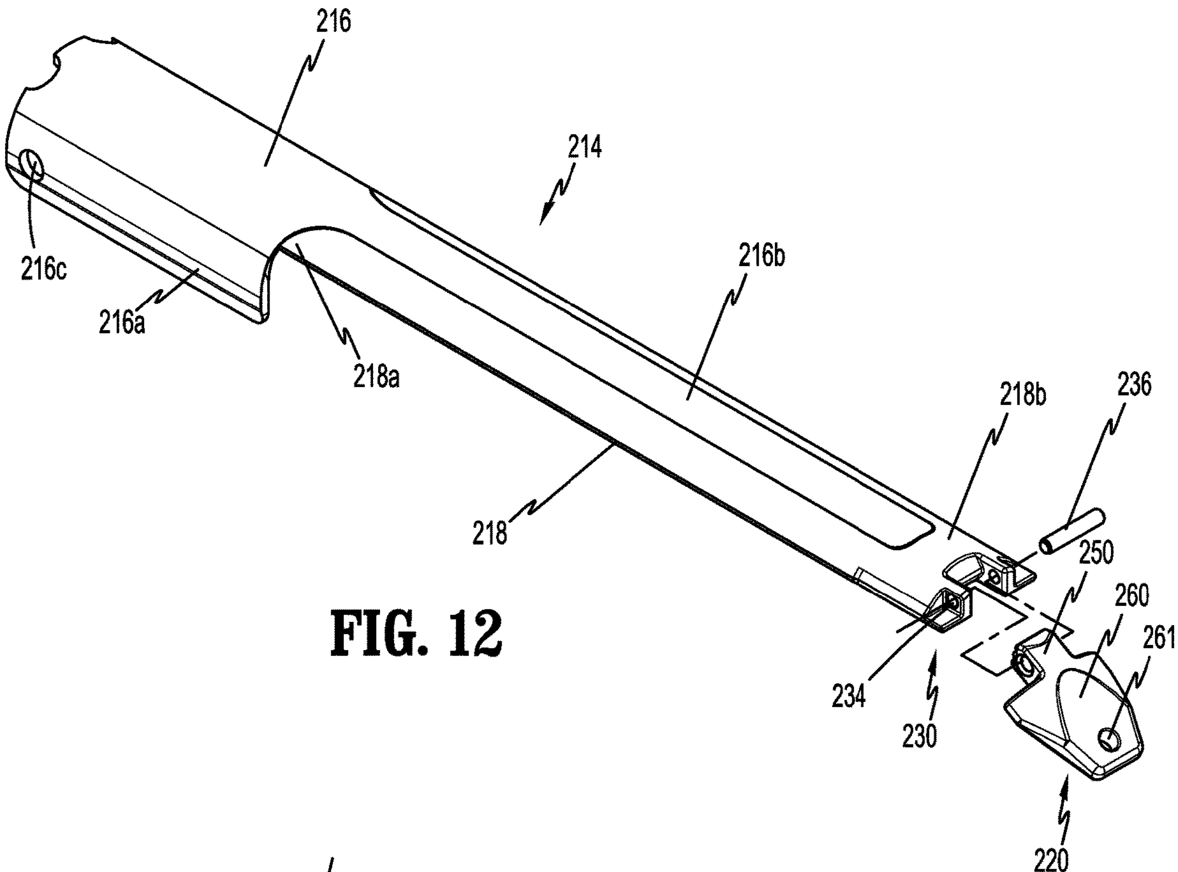


FIG. 12

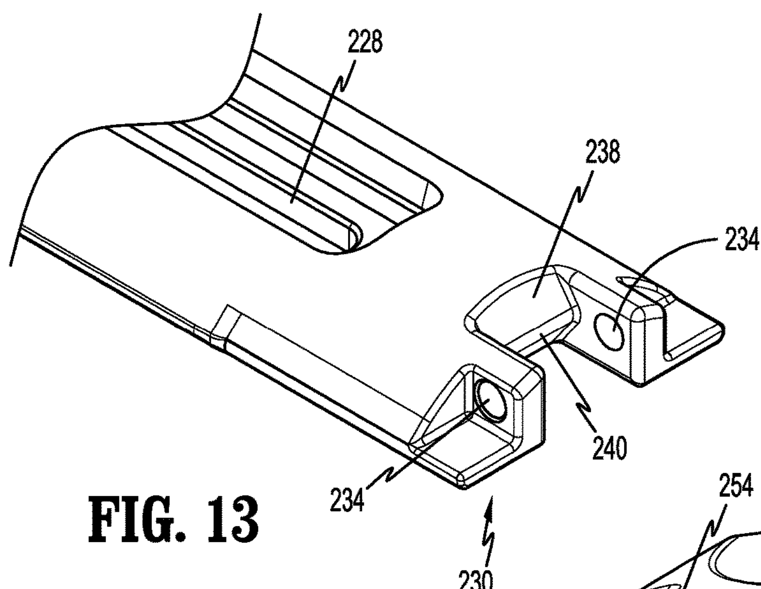


FIG. 13

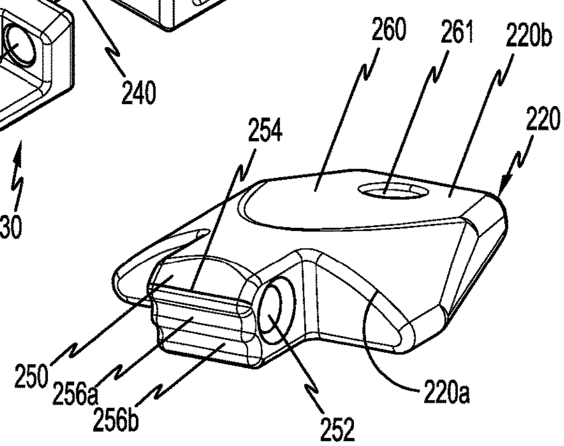
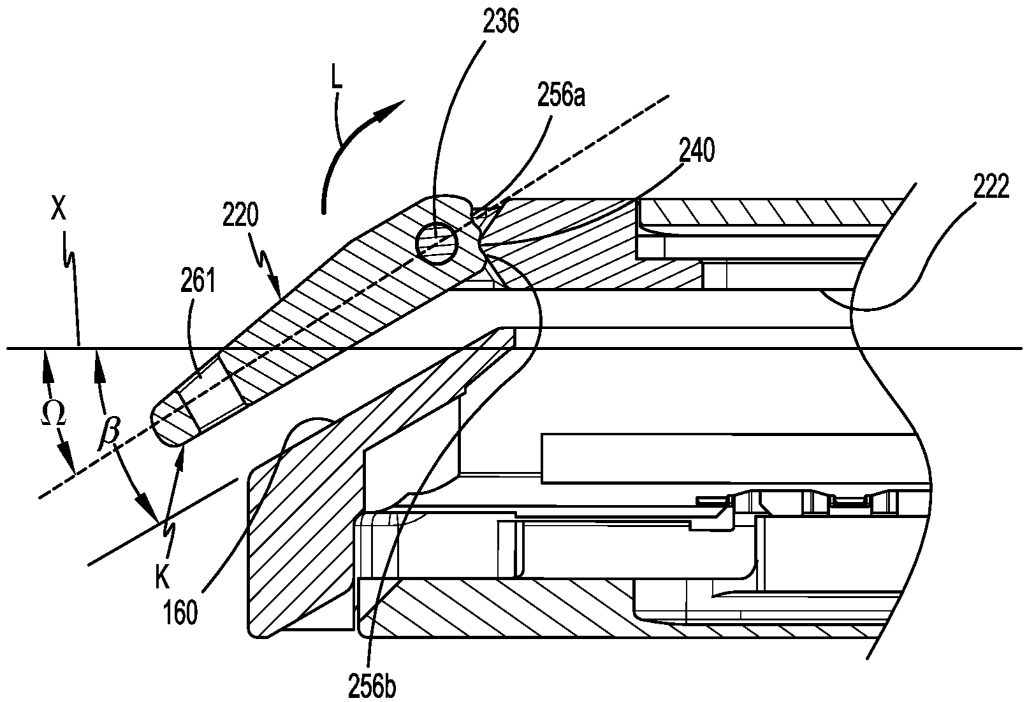


FIG. 14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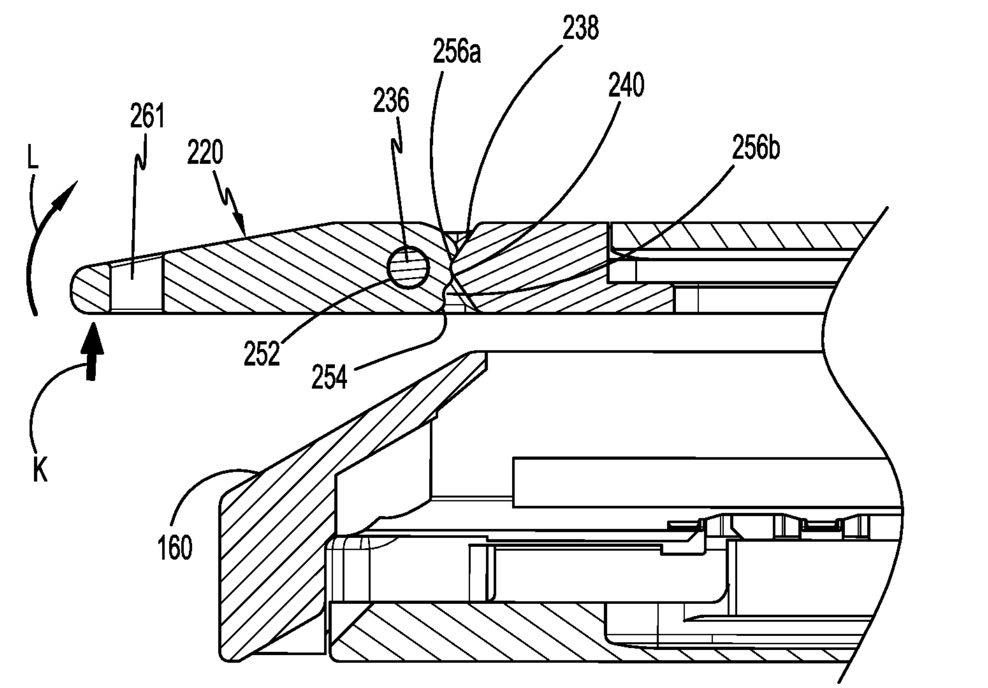
222
X
220
261
Ω
β
K
160
256b

FIG. 15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L
252
254
K
160

FIG. 16

SURGICAL STAPLER WITH UNIVERSAL TIP RELOAD

FIELD

The disclosure is directed to surgical stapling devices and, more particularly, to endoscopic surgical stapling devices.

BACKGROUND

Surgical stapling devices for performing surgical procedures endoscopically are well known. Such devices are available in a variety of different configurations, e.g., linear, curved, circular, etc., and are suitable for use in a variety of different procedures. Linear surgical stapling devices include a tool assembly having an anvil and a staple cartridge that are pivotably coupled to each other at their proximal ends between open and clamped positions. In order to better navigate the tool assembly to a surgical site endoscopically, the tool assembly may include a dissector tip that extends from a distal end of the tool assembly. Typically, the dissector tip is supported on the tool assembly and is configured to separate target tissue from body tissue to facilitate placement of the tool assembly about the target tissue. During some surgical procedures, it is desirable to have an angled dissector tip whereas in other surgical procedures it is desirable to have a linear dissector tip.

A continuing need exists in the art for a surgical stapling device that is better suited to access to a variety of surgical sites.

SUMMARY

One aspect of this disclosure is directed a surgical stapler including an elongate body and a tool assembly. The elongate body has a proximal portion and a distal portion. The tool assembly defines a longitudinal axis and includes a cartridge assembly, an anvil assembly, and a dissector tip. The anvil assembly has a proximal portion and a distal portion and is coupled to the cartridge assembly such that the tool assembly is moveable from an open position to a clamped position. The dissector tip includes a body having a proximal end and a distal end. The body has a thickness that decreases towards the distal end. The body is movably coupled to the anvil assembly for movement between a first configuration substantially aligned with the longitudinal axis and a second configuration defining an acute angle with the longitudinal axis.

In aspects of the disclosure, the dissector tip includes a tip beam having a proximal portion and a deformable distal beam portion, wherein the deformable distal beam portion is bendable between the first and second configurations.

In some aspects of the disclosure, the distal portion of the anvil assembly includes a bracket and the proximal portion of the tip beam is secured to the bracket.

In certain aspects of the disclosure, the proximal portion of the tip beam is secured to the bracket of the anvil assembly by welding or other means (mechanical snap, adhesive, pin, etc).

In aspects of the disclosure, the dissector tip includes a tip beam cover that is supported on the deformable distal beam portion and has an atraumatic configuration.

In some aspects of the disclosure, the tip beam cover is formed of a biocompatible material selected from the group consisting of polymeric materials, metals, ceramics, and elastomeric materials.

In certain aspects of the disclosure, the dissector tip includes a body having a proximal portion and a distal portion, wherein the proximal portion includes a hinge portion that is pivotably coupled to the distal portion of the anvil assembly.

In aspects of the disclosure, the distal portion of the anvil assembly includes a clevis and the hinge portion is coupled to the clevis by a clevis pin.

In some aspects of the disclosure, the tool assembly includes retaining structure configured to retain the dissector tip in one of the first and second configurations.

In certain aspects of the disclosure, the retaining structure includes a protrusion formed on one of the distal portion of the anvil assembly or the proximal portion of the dissector tip and a plurality of recesses formed on the other of the distal portion of the anvil assembly or the proximal portion of the dissector tip.

In one aspect of the disclosure, the dissector tip has a width that decreases towards the distal end of the dissector tip.

Another aspect of the disclosure is directed to a tool assembly including a cartridge assembly, an anvil assembly, and a dissector tip. The anvil assembly includes a proximal portion and a distal portion and is coupled to the cartridge assembly such that the tool assembly is moveable from an open position to a clamped position. The tool assembly defines a longitudinal axis. The dissector tip includes a body having a proximal end and a distal end and has a thickness that decreases towards the distal end. The body is movably coupled to the anvil assembly for movement between a first configuration substantially aligned with the longitudinal axis and a second configuration defining an acute angle with the longitudinal axis.

Another aspect of the disclosure is directed to a tool assembly defining a longitudinal axis and including a cartridge assembly, an anvil assembly, and a dissector tip. The anvil assembly includes a base portion and an anvil portion. The base portion is coupled to the cartridge assembly such that the tool assembly is moveable from an open position to a clamped position. The anvil portion is secured to the base portion and includes a plurality of staple deforming pockets. The anvil portion includes a distal portion supporting a bracket. The dissector tip includes a body having a tapered distal surface and a thickness that decreases in the distal direction. The body is movably coupled to the anvil assembly for movement between a first configuration substantially aligned with the longitudinal axis and a second configuration defining an acute angle with the longitudinal axis. The body of the dissector tip includes a tip beam having a proximal portion and a deformable distal beam portion. The proximal portion of the dissector tip is secured to the bracket of the anvil assembly and the deformable distal beam portion is bendable between the first and second configurations.

Other features of the disclosure will be appreciated from the following description.

BRIEF DESCRIPTION OF THE DRAWINGS

Various aspects of the disclosed surgical stapling device are described herein below with reference to the drawings, wherein:

FIG. **1** is a side perspective view of exemplary aspects of the disclosed surgical stapling device including a reload assembly with a universal tip;

FIG. **2** is an enlarged view of the indicated area of detail shown in FIG. **1**;

FIG. **3** is a perspective bottom view of an anvil assembly of the reload assembly of the surgical stapling device shown in FIG. **1**;

FIG. **4** is an exploded, perspective view of the anvil assembly shown in FIG. **3**;

FIG. **5** is an enlarged cutaway, perspective view of a distal portion of an anvil body and tip beam of the anvil assembly shown in FIG. **4**;

FIG. **6** is a side perspective view of a tool assembly of the reload assembly shown in FIG. **1** in a clamped position;

FIG. **7** is a cross-sectional view taken along section line **7-7** of FIG. **6** with the anvil tip in a first position;

FIG. **8** is a cross-sectional view taken along section line **7-7** of FIG. **6** with the anvil tip in a second position;

FIG. **9** is a cross-sectional view taken along section line **7-7** of FIG. **6** with the anvil tip in a third position;

FIG. **10** is a side view of other exemplary aspects of a tool assembly of the reload assembly of the stapling device shown in FIG. **1** with the tool assembly in the clamped position;

FIG. **11** is an enlarged side perspective view of a distal end of the tool assembly shown in FIG. **10** with the tool assembly in the clamped position;

FIG. **12** is a side perspective, exploded view of the anvil assembly of the tool assembly shown in FIG. **10**;

FIG. **13** is an enlarged view of the distal end of the anvil base of the anvil assembly shown in FIG. **10**;

FIG. **14** is a perspective view from the proximal end of the dissector tip of the anvil assembly shown in FIG. **11**;

FIG. **15** is a cross-sectional view taken along section line **15-15** of FIG. **11** with the dissector tip in a first position; and

FIG. **16** is a cross-sectional view taken along section line **15-15** of FIG. **11** with the dissector tip in a second position.

DETAILED DESCRIPTION

The disclosed surgical stapling device will now be described in detail with reference to the drawings in which like reference numerals designate identical or corresponding elements in each of the several views. However, it is to be understood that the aspects of the disclosure described herein are merely exemplary of the disclosure and may be embodied in various forms. Well-known functions or constructions are not described in detail to avoid obscuring the disclosure in unnecessary detail. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the disclosure in virtually any appropriately detailed structure.

In this description, the term "proximal" is used generally to refer to that portion of the device that is closer to a clinician, while the term "distal" is used generally to refer to that portion of the device that is farther from the clinician. In addition, the term "endoscopic" is used generally to refer to endoscopic, laparoscopic, arthroscopic, and/or any other procedure conducted through a small diameter incision or cannula. Further, the term "clinician" is used generally to refer to medical personnel including doctors, nurses, and support personnel. As used herein, the terms "parallel" and "aligned" are understood to include relative configurations that are substantially parallel, and substantially aligned, i.e., up to about + or −10 degrees from true parallel or true alignment.

The disclosed surgical stapling device includes a tool assembly that supports a dissector tip that is adjustable to allow a clinician to position the dissector tip in a configuration that facilitates easy access to a surgical site. Some aspects of the disclosure are directed to a dissector tip that is bendable between a first orientation in which an axis defined by dissector tip is parallel to a longitudinal axis of the tool assembly and a second position in which the axis defined by the dissector tip defines an acute angle with the longitudinal axis of the tool assembly. Other aspects of the disclosure are directed to a dissector tip that is pivotable between a first orientation in which the axis defined by the dissector tip is parallel to a longitudinal axis of the tool assembly and a second position in which the axis defined by the dissector tip defines an acute angle with the longitudinal axis of the tool assembly.

FIG. **1** illustrates the disclosed surgical stapling device **10** including a handle assembly **12**, an elongate body **14**, and a tool assembly **100**. Although not described in detail herein, the tool assembly **100** can form part of a reload assembly **16** that is releasably coupled to the elongate body **14** and includes a proximal body portion **16*a*** and the tool assembly **100**. Alternately, the tool assembly **100** can be fixedly secured to a distal portion of the elongate body **14**. The handle assembly **12** includes a hand grip **18**, a plurality of actuator buttons **20**, and a rotation knob **22**. The rotation knob **22** facilitates rotation of the elongate body **14** and the tool assembly **100** in relation to the handle assembly **12**. The actuator buttons **20** control operation of the various functions of the stapling device **10** including approximation, firing and cutting. Although the stapling device **10** is illustrated as an electrically powered stapling device, it is envisioned that the disclosed tool assembly **100** would also be suitable for use with a manually powered surgical stapling device. U.S. Pat. No. 9,055,943 (the '943 Patent) discloses a surgical stapling device including a powered handle assembly and U.S. Pat. No. 6,241,139 (the '139 Patent) discloses a manually actuated handle assembly. The tool assembly **100** includes a cartridge assembly **112**, an anvil assembly **114**, and a dissector tip **120**. For a more detailed description of the cartridge assembly **112** of the tool assembly **100**, see, e.g., the '943 and '139 Patents.

FIGS. **2-7** illustrate the anvil assembly **114** of the tool assembly **100** which defines a longitudinal axis "X" and includes a base portion **116** and an anvil portion **118**. The base portion **116** includes a proximal mounting portion **116*a*** and a longitudinal rib **116*b*** that extends distally from the proximal mounting portion **116*a***. The proximal mounting portion **116*a*** includes spaced through bores **116*c*** that receive pivot members **117** (FIG. **1**) that pivotably couple the cartridge assembly **112** to the anvil assembly **114**. The anvil portion **118** is secured to the base portion **116** by, e.g., welding, and includes an upper surface **122**, a proximal portion **118*a***, and a distal portion **118*b***. The upper surface **122** of the anvil portion **118** defines a plurality of rows of staple deforming pockets **126** that are positioned on opposite sides of a knife slot **128**. The distal portion **118*b*** of the anvil portion **118** includes a support plate or bracket **130**.

The dissector tip **120** includes a tip beam **140** and a tip beam cover **142**. The tip beam **140** includes a body having a proximal portion **144** and a deformable distal beam portion **146**. The tip beam **140** is secured to the support bracket **130** on the distal portion **118*b*** of the anvil portion **118** and is formed from a substantially rigid material that can be bent and shaped to a desired configuration such as a metal. The tip beam cover **142** has an atraumatic configuration and is formed of any suitable biocompatible material including polymeric materials, metals, ceramics, elastomeric materials, etc. In certain aspects of the disclosure, the tip beam cover **142** has rounded edges **156** and a tapered distal tip **158**

(FIG. **4**) that decreases in width in the distal direction. The tip beam cover **142** also has a thickness that decreases in the distal direction such that the cover **142** is thinnest at its distal end. The reduced thickness allows for the dissector tip **120** to more easily navigate through and dissect tissue. Other tip beam cover configurations are envisioned.

FIGS. **8** and **9** illustrate the distal portion of the tool assembly **100** as the dissector tip **120** is moved between a first configuration (FIG. **8**) in which the dissector tip **120** extends in a direction substantially parallel to the longitudinal axis "X" of the tool assembly **100** and a second configuration (FIG. **9**) in which the dissector tip **120** of the tool assembly **100** extends in a direction to define an acute angle Ω with the longitudinal axis "X" of the tool assembly **100**. In certain aspects of the disclosure, a distal portion **112***a* of the cartridge assembly **112** defines a tissue guide surface **160** that defines an angle θ with the longitudinal axis "X" of the tool assembly **100**. In order to move the dissector tip **120** between the first and second configurations, the tip beam **140** and the tip beam cover **142** are manually grasped by a clinician and bent or deformed in the direction of arrows "A" or "B" in FIGS. **8** and **9**, respectively, to a desired configuration best suited to perform a certain surgical procedure e.g., in which angle Ω is from about zero to about 45 degrees. The tip beam **140** is formed of a material that will retain its deformed configuration as the dissector tip **120** is used to separate target tissue from body tissue as the tool assembly **100** is advanced endoscopically to a surgical site.

FIGS. **10-14** illustrate other exemplary aspects of a tool assembly **200** according to the disclosure. The tool assembly **200** includes an anvil assembly shown generally as anvil assembly **214**. The anvil assembly **214** is similar to the anvil assembly **114** but includes modifications to the dissector tip **120**. Only those modifications to the dissector tip shown generally as dissector tip **220** will be described in detail herein.

The anvil assembly **214** includes a base portion **216**, an anvil portion **218**, and the dissector tip **220**. The base portion **216** includes a proximal mounting portion **216***a* and a longitudinal rib **216***b* that extends distally from the proximal mounting portion **216***a* towards a distal end of the anvil portion **218**. The proximal mounting portion **216***a* includes spaced through bores **216***c* that receive pivot members **117** (FIG. **1**) that pivotably couple the cartridge assembly **112** (FIG. **1**) to the anvil assembly **214**. The anvil portion **218** is secured to the base portion **216** by, e.g., welding, and includes an upper surface **222** (FIG. **15**), a proximal portion **218***a*, and a distal portion **218***b*. The upper surface **222** of the anvil portion **118** defines a plurality of rows of staple deforming pockets (not shown) that are positioned on opposite sides of a knife slot **228** (FIG. **13**). The distal portion **218***b* of the anvil portion **118** includes a clevis **230** that defines spaced openings **234** that receive a clevis pin **236**. The distal portion **218***b* of the anvil portion **218** includes a distal face **238** that has a protrusion **240**.

The dissector tip **220** has a body including a proximal portion **220***a* and a distal portion **220***b*. The proximal portion **220***a* includes a hinge portion **250** that defines a through bore **252** that receives the clevis pin **236** to pivotably secure the dissector tip **220** to the distal end portion **218***b* of the anvil portion **218**. The dissector tip **220** has a proximal face **254** that defines spaced recesses **256***a* and **256***b* that are positioned to receive the protrusion **240** on the distal face **238** of the anvil portion **218** (FIG. **15**). The dissector tip **220** has a tapered distal face **260** in the distal direction (FIG. **15**) such that the distal end of the dissector tip **220** has a height that decreases in the distal direction. In addition, the width of the dissector tip **220** may also decrease in the distal direction (FIG. **11**.) The dissector tip **220** has a through bore **261** that extends through the tapered distal face **260**. The through bore **261** allows a clinician to lasso or grasp the dissector tip **220** with a suture to allow the clinician to more accurately move and position the tool assembly **200** within a body cavity.

FIGS. **15** and **16** illustrate the dissector tip **220** in a first configuration (FIG. **15**) and in a second configuration (FIG. **16**). In the first configuration (FIG. **15**), the dissector tip **220** defines a longitudinal axis that defines an acute angle Ω with the longitudinal axis "X" of the tool assembly **200**. As discussed above, it is envisioned that angle Ω can be substantially the same as angle θ defined by the tissue guide surface **160** of the cartridge assembly **112**. In the second configuration, the dissector tip **220** defines a longitudinal axis that is substantially parallel to the longitudinal axis "X of the tool assembly **200**. In order to move the dissector tip **220** between the first and second configurations, a clinician can apply a force to the dissector tip **220** to disengage the protrusion **240** from one of the respective spaced recesses **256***a* and **256***b* and pivot the dissector tip **220** about the clevis pin **236** to position the protrusion **240** in the other recess **256***a* or **256***b* of the dissector tip **240**. For example, when the dissector tip **220** is moved from the first configuration (FIG. **15**) to the second configuration (FIG. **16**), the clinician applies a force on the dissector tip **220** in the direction indicated by arrows "K" to pivot the dissector tip **220** about the clevis pin **236** in the direction indicated by arrow "L" to remove the protrusion **240** from the recess **256***b* and move the protrusion into the recess **256***a*. Although only two configurations are shown in the figures, it is envisioned that the dissector tip **220** may be moved to multiple positions. In that respect, it is envisioned that two or more spaced recesses **256** can be provided to receive the protrusion **240** to retain the dissector tip **220** in multiple different angular positions in relation to the longitudinal axis of the tool assembly **200**.

Although the drawings only show retaining structure including a protrusion and a plurality of recesses, it is also envisioned that a variety of different types of retention structures can be provided to releasably retain the dissector tip in different angular positions. Further, the retaining structure can be supported on or integrally formed with either or both of the dissection tip **220** and the anvil assembly **214**.

Persons skilled in the art will understand that the devices and methods specifically described herein and illustrated in the accompanying drawings are non-limiting exemplary aspects of the disclosure. It is envisioned that the elements and features illustrated or described in connection with one exemplary embodiment may be combined with the elements and features of another without departing from the scope of the disclosure. As well, one skilled in the art will appreciate further features and advantages of the disclosure based on the above-described aspects of the disclosure. Accordingly, the disclosure is not to be limited by what has been particularly shown and described, except as indicated by the appended claims.

What is claimed is:

1. A surgical stapler comprising:

an elongate body having a proximal portion and a distal portion; and

a tool assembly defining a longitudinal axis and including:

a cartridge assembly;

an anvil assembly including a proximal portion and a distal portion, the distal portion of the anvil assembly having a support bracket, the anvil assembly coupled to the cartridge assembly such that the tool assembly is moveable from an open position to a clamped position; and

a dissector tip including a body having a proximal end and a distal end, the body having a thickness that decreases towards the distal end, the body being movably coupled to the anvil assembly for movement between a first configuration substantially aligned with the longitudinal axis and a second configuration defining an acute angle with the longitudinal axis, wherein the dissector tip includes a tip beam and a tip beam cover, the tip beam including a proximal portion and a deformable distal beam portion, the proximal portion of the tip beam fixedly secured to the support bracket, the tip beam cover received about the tip beam and the support bracket and having an atraumatic configuration, the deformable distal beam portion being formed of a material that is manually bendable by a clinician to any desired configuration including and between the first and second configurations, the material having properties to retain the deformable distal beam portion in the desired configuration absent an external force applied to the dissector tip.

2. The surgical stapler of claim **1**, wherein the proximal portion of the tip beam is secured to the support bracket of the anvil assembly by welding.

3. The surgical stapler of claim **1**, wherein the tip beam cover is formed of a biocompatible material selected from the group consisting of polymeric materials, metals, ceramics, and elastomeric materials.

4. The surgical stapler of claim **1**, wherein the dissector tip has a width that decreases towards the distal end of the dissector tip.

5. A tool assembly comprising:

a cartridge assembly;

an anvil assembly including a proximal portion and a distal portion, the distal portion of the anvil assembly having a support bracket, the anvil assembly coupled to the cartridge assembly such that the tool assembly is moveable from an open position to a clamped position, the tool assembly defining a longitudinal axis; and

a dissector tip including a body having a proximal end and a distal end, the body having a thickness that decreases towards the distal end, the body being movably coupled to the anvil assembly for movement between a first configuration substantially aligned with the longitudinal axis and a second configuration defining an acute angle with the longitudinal axis, wherein the dissector tip includes a tip beam and a tip beam cover, the tip beam including a proximal portion and a deformable distal beam portion, the proximal portion of the tip beam fixedly secured to the support bracket, the tip beam cover received about the tip beam and the support bracket and having an atraumatic configuration, the deformable distal beam portion being formed of a material that is manually bendable by a clinician to any desired configuration including and between the first and second configurations, the material having properties to retain the tip beam portion in the desired configuration absent an external force applied to the dissector tip.

6. A tool assembly comprising:

a cartridge assembly;

an anvil assembly including a proximal portion and a distal portion, the anvil assembly including a base portion and an anvil portion, the base portion coupled to the cartridge assembly such that the tool assembly is moveable from an open position to a clamped position, the anvil portion secured to the base portion and including a plurality of staple deforming pockets, the anvil portion including a distal portion including a support bracket, the tool assembly defining a longitudinal axis; and

a dissector tip including a body having a tapered distal surface and a thickness that decreases in the distal direction, the body being movably coupled to the anvil assembly for movement between a first configuration substantially aligned with the longitudinal axis and a second configuration defining an acute angle with the longitudinal axis, the dissector tip including a tip beam and a tip beam cover, the tip beam having a proximal portion and a deformable distal beam portion, the proximal portion of the tip beam fixedly secured to the support bracket, the tip beam cover received about the tip beam and the support bracket and having an atraumatic configuration, the proximal portion of the dissector tip secured to the bracket and the deformable distal beam portion formed of a material that is manually bendable by a clinician to any desired configuration including and between the first and second configurations, the material having properties to retain the distal beam portion in the desired configuration absent an external force applied to the dissector tip.

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